

# Assignment 17: User Onboarding Validation Module

Source Code & Unit Test Submission Document

## Complete Script File: assignment\_17.py

```
import re
import pytest

# =====
# 1. CUSTOM EXCEPTION CLASSES
# =====

class InvalidEmailError(ValueError):
    """Exception raised when the provided email format is invalid."""
    def __init__(self, email):
        self.email = email
        super().__init__(f"Invalid email format: {email}")

class UnderageError(Exception):
    """Exception raised when the user is under the required age limit."""
    def __init__(self, age):
        self.age = age
        super().__init__(f"User must be at least 18 years old. Age given: {age}")

# =====
# 2. CORE REGISTRATION SERVICE
# =====

class RegistrationService:
    # Industry-standard email validation regex pattern
    EMAIL_PATTERN = r"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"

    def register_user(self, email: str, age: int) -> bool:
        """Validates user onboarding information."""
        assert isinstance(age, int), "Age must be an integer"

        if email is None or not email.strip():
            raise InvalidEmailError(email)

        if not re.match(self.EMAIL_PATTERN, email):
            raise InvalidEmailError(email)

        if age < 18:
            raise UnderageError(age)

        print("Registration successful")
        return True
```

```

# =====
# 3. PYTEST UNIT TEST SUITE
# =====

@pytest.fixture
def service():
    """Fixture to provide a clean instance of RegistrationService for each test."""
    return RegistrationService()

def test_valid_registration(service):
    assert service.register_user("alice@example.com", 25) is True

def test_invalid_email(service):
    with pytest.raises(InvalidEmailError):
        service.register_user("wrong-email", 20)

def test_underage_user(service):
    with pytest.raises(UnderageError):
        service.register_user("bob@example.com", 16)

# =====
# 4. LOCAL DRIVER CODE
# =====
if __name__ == "__main__":
    # This block runs when executing the file directly via python assignment_17.py
    print("--- Running local driver execution check ---")
    service_instance = RegistrationService()
    try:
        service_instance.register_user("john@example.com", 22)
    except (InvalidEmailError, UnderageError) as e:
        print(f"Caught expected error: {e}")

```